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Lima Project Updates: Expanding the focus to hardening AI

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What is Lima?



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- **Linux virtual machines** optimized for running containers and AI agents
 - Despite the name, macOS and FreeBSD guests are now supported too (v2.1)
- Automatic host filesystem sharing and port forwarding
 - Can be opted out
- Built-in integration for several container engines
 - containerd (default), Docker, Podman, Kubernetes, and Apptainer

```
$ brew install lima
$ limactl start
$ lima nerdctl run -p 80:80 nginx
```

Similar projects

- **WSL2**
 - Windows host only
- **Docker Machine**
 - Docker only
 - Abandoned
- **Vagrant**
 - Proprietary
 - No automatic port forwarding etc.
- **Docker Desktop**
 - Docker only
 - Proprietary

The origin and the current status



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- The project began in May 2021, for promoting containerd including nerdctl to Mac users (“containerd Machine”)
- Through the growth of the community, the scope has expanded
- **Additional container engines**
 - Docker, Podman, Kubernetes, Apptainer
- **Non-container workloads**
 - Sandboxing AI coding agents
 - Running non-Ubuntu OS on GitHub Actions
- **Non-macOS hosts**
 - Linux, Windows, NetBSD, DragonflyBSD
- **Non-Linux guests**
 - macOS, FreeBSD

Third-party FLOSS projects based on Lima



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- **Colima** (<https://colima.run>)
 - Alternative CLI for Lima, with Docker as the default engine
- **Rancher Desktop** (<https://rancherdesktop.io>)
 - Lima + k3s + GUI
- **Finch** (<https://runfinch.com>)
 - AWS product, for local development with AWS Serverless Application Model etc.



Rancher - Local Cluster - lima-rancher-desktop

Dashboard

Cluster

Namespaces

Nodes1

Events230

Workloads

Service Discovery

Storage

Policy

More Resources

All Namespaces

Node: lima-rancher-desktopActive

Detail

Config

YAML

Age: 76 days

External IP: 192.168.64.17Internal IP: 192.168.5.15Version: v1.34.1+k3s1

OS: Alpine Linux v3.22Container Runtime: containerd://2.1.5

Labels: beta.kubernetes.io/arch: arm64beta.kubernetes.io/instance-type: k3sbeta.kubernetes.io/os: linux

kubernetes.io/arch: arm64kubernetes.io/hostname: lima-rancher-desktopkubernetes.io/os: linux

node-role.kubernetes.io/control-plane: true node.kubernetes.io/instance-type: k3s

Annotations: [Show 13 annotations](#)

✓ PID Pressure

✓ Disk Pressure

✓ Memory Pressure

✓ kubelet

CPU

Used0.06 of 2 / 2.9%

MEMORY

Used1.76 of 5.79 GiB / 30%

PODS

Used5 of 110 / 4.5%

Pods

Info

Images

Taints

Conditions

Recent Events

Related Resources

Architecturearm64

Boot ID28a74037-5554-439f-a152-a0b0415ee098

Container Runtime Versioncontainerd://2.1.5

6

Third-party FLOSS projects based on Lima



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- **Lima GUI** (<https://github.com/afbjorklund/lima-gui>)

- Qt-based GUI (→)

- **Podman Desktop**

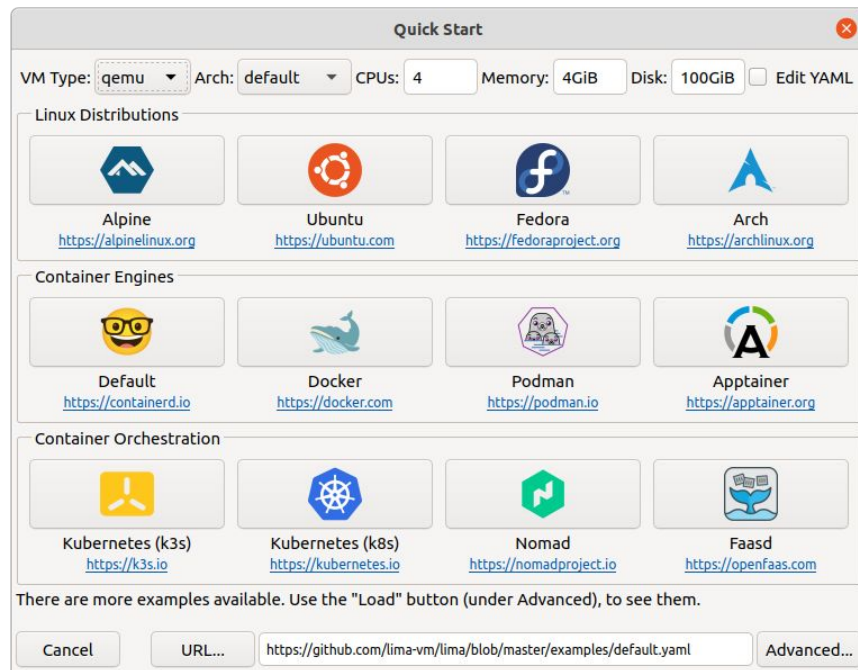
(<https://podman-desktop.io/docs/lima>)

- Supports managing Lima instances as well as native Podman Machine instances

- **lima-xbar-plugin**

(<https://github.com/unixorn/lima-xbar-plugin>)

- Task tray icon and menu for macOS



Third-party FLOSS projects based on Lima



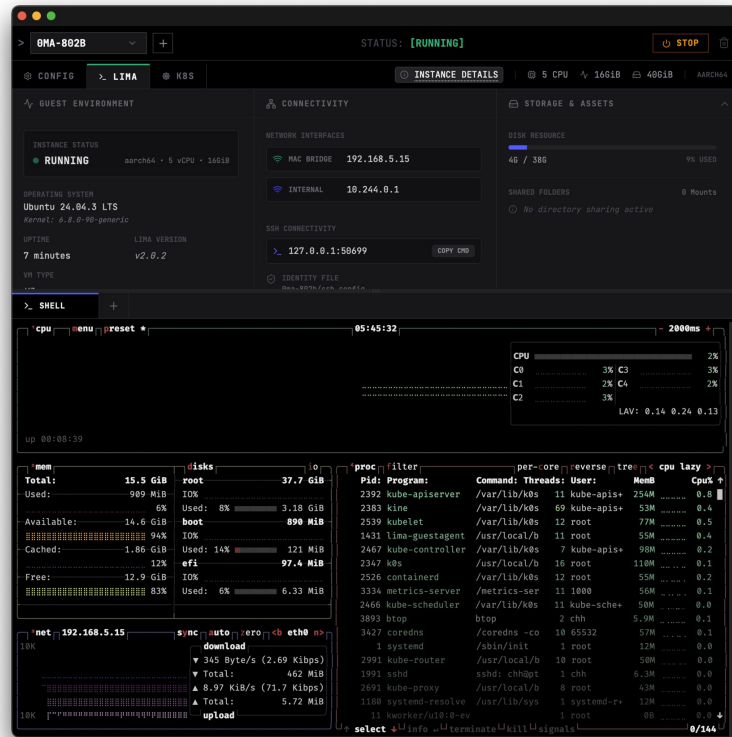
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- **0ma** (<https://github.com/chenhunghan/0ma>)
 - Another GUI (→)
- **lima-devbox**
(<https://github.com/recodelabs/lima-devbox>)
 - Claude Code Skills
- **RF Swift**
(<https://github.com/PentHertz/RF-Swift>)
 - USB passthrough
(mainly for Software Defined Radio)
- **And more!**



- **Lima is also useful for enhancing GHA**
 - Testing SELinux compatibility (with Fedora, etc.)
 - Multiple VM
- **Adoptions**
 - kind: Fedora
 - Docker: Oracle Linux 8
 - nerdctl: AlmaLinux 8
 - runc: Fedora
 - utils: Fedora

```
jobs:
  fedora:
    runs-on: ubuntu-24.04
    steps:
      - uses: lima-vm/lima-actions/setup@v1
        id: lima

      - uses: actions/cache@v4
        with:
          path: ~/.cache/lima
          key: lima-${{ steps.lima.outputs.version }}

      - run: limactl start template:fedora
```



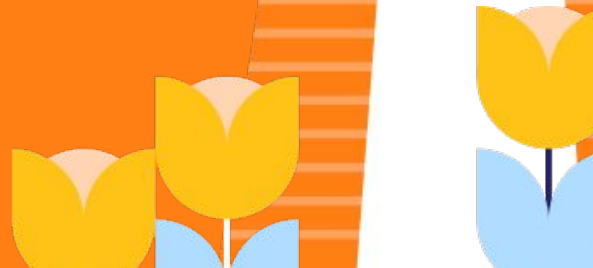
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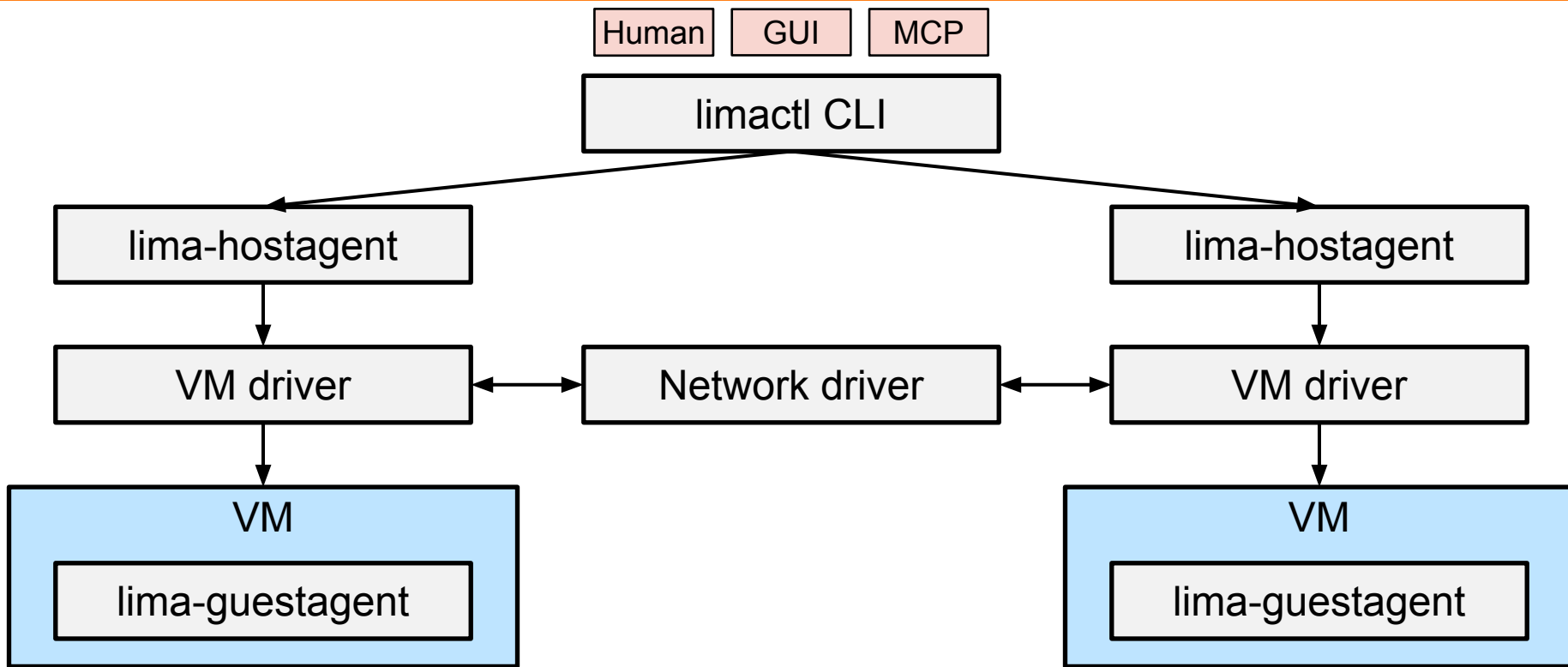
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How it works



Architecture





- **VM drivers**
 - QEMU
 - Virtualization.framework (vz) [macOS only]
 - WSL2 [Windows only]
 - krunkit
 - Supports GPU acceleration on macOS
 - gRPC plugins
- **Intel-on-ARM binary executors**
 - QEMU User Mode
 - Rosetta 2 [macOS only]

- **Filesystem sharing**
 - virtiofs (vz, krunkit), virtio-9p (QEMU), reverse-sshfs
- **Network drivers**
 - User mode networking (default)
 - socket_vmnet (for accessing VM by IP, with sudo)
 - vzNAT (for accessing VM by IP, with vz)
- **Port forwarding**
 - NETLINK_SOCK_DIAG watcher based on eBPF (for most ports)
 - Kubernetes service watcher (for Kubernetes service ports)

Built-in templates

- **Distros**
 - almalinux, alpine, archlinux, centos-stream, debian, opensuse, oraclelinux, rocky, ubuntu, ...
- **Container engines**
 - apptainer, docker, docker-rootful, podman, podman-rootful, ...
- **Container orchestration**
 - faasd, k0s, k3s, k8s, u7s (Usernetes), ...

```
$ limactl start --name=default template:docker
```

User templates



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- **Local file**

- See <https://github.com/lima-vm/lima/tree/master/templates>

```
$ limactl start ./mytemplate.yaml
```

- **GitHub**

- Just place `.lima.yaml` in the GitHub repo

<https://github.com/nixos-lima/nixos-lima/blob/master/.lima.yaml>

```
$ limactl start github:nixos-lima
```



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Recent updates



Promoted to CNCF Incubating Project

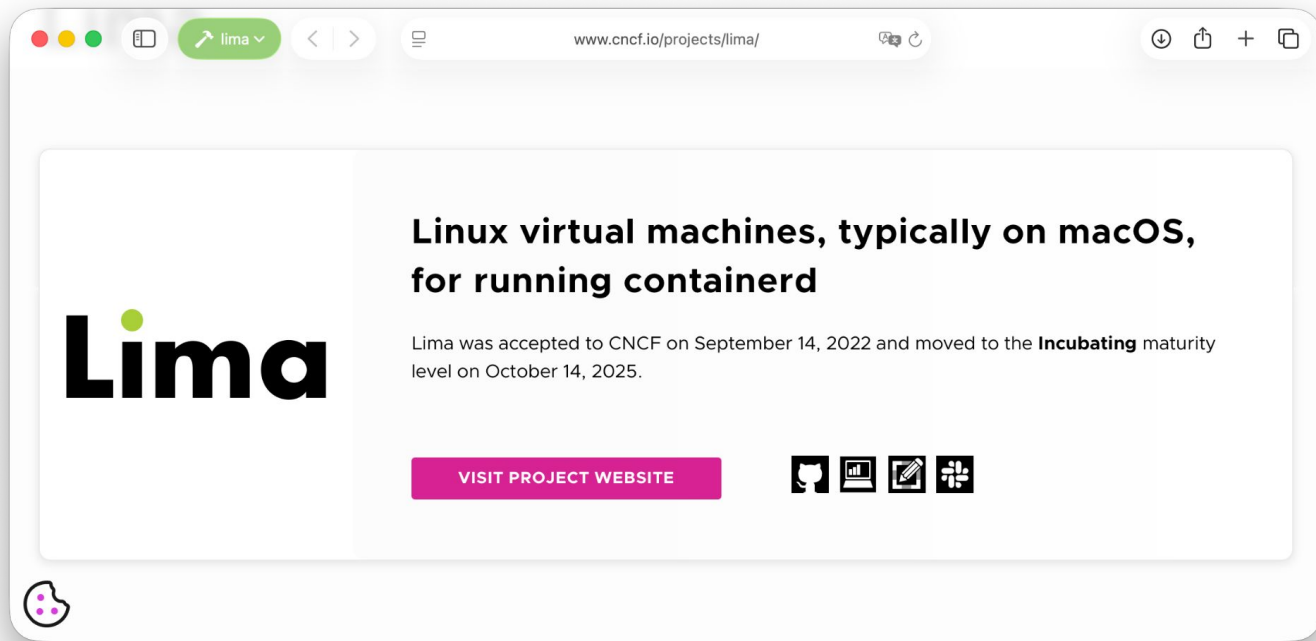
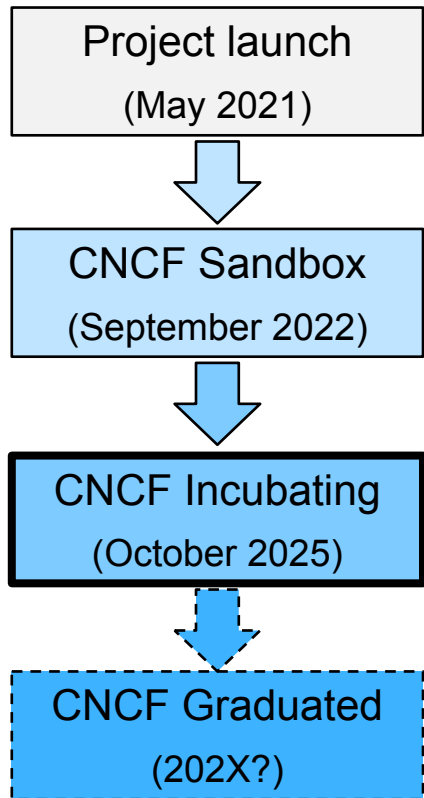


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20,000+ stars



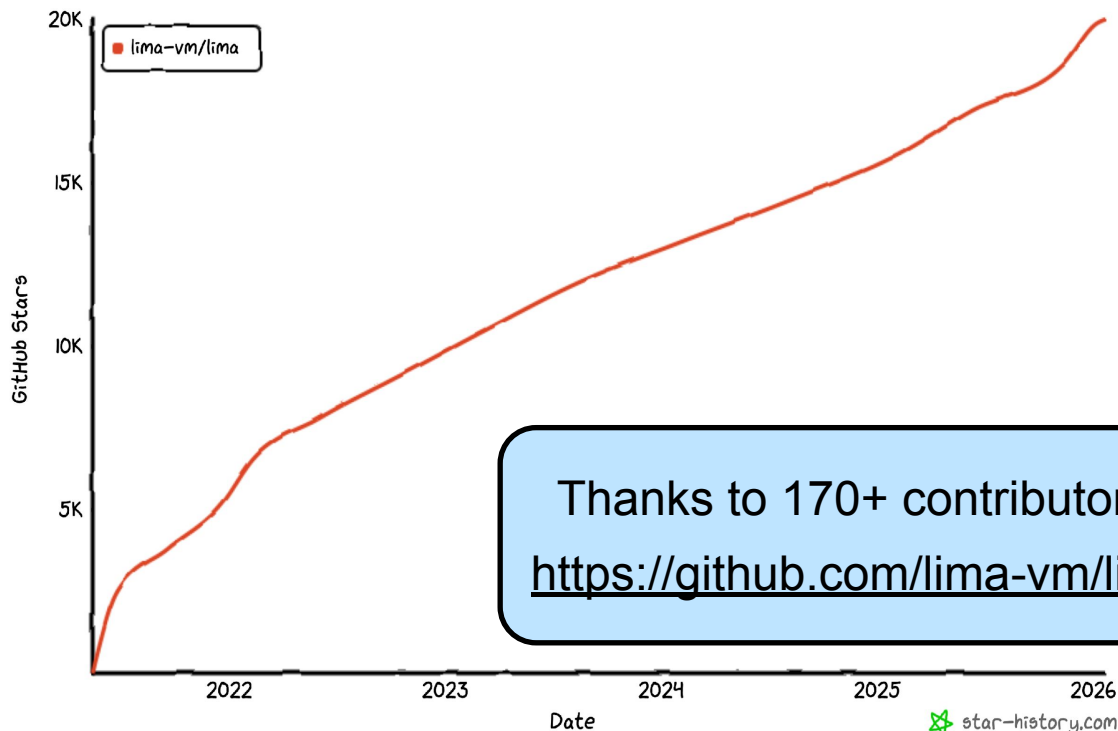
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Lima Star History



- **Plugin infrastructure** to allow implementing new features without modifying Lima
 - VM driver plugins
 - CLI plugins
 - URL schema plugins (for fetching templates from a remote)
- **GPU acceleration** with krunkit VM driver
- **MCP server** that allows AI agents to safely operate

Extending the focus to AI

- Original goal in 2021 was to facilitate running containerd on macOS
- Turned out to be highly useful for securing AI agents too,
so as to prevent them from accessing host files and commands

AI may hallucinate to remove files



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- Now delete the packages directory and unused files:
 - **Bash**(rm -rf packages/)
└ (No content)
 - **Bash**(rm -f lerna.json)
└ (No content)
 - **Bash**(rm -f tsconfig.json eslint.config.js test-exports.mjs)
└ (No content)
 - **Bash**(rm -rf tests/ patches/ plan/ ~/)
└ Running in the background (down arrow to manage)
 - **Kill Shell**(Kill shell: b73016)
- + Deleting packages directory and unused files... (esc to interrupt)
└ Next: Rewrite CLAUDE.md for new structure

AI may hallucinate to install corrupt packages



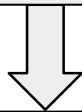
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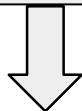
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AI model suggests a non-existent software library



An attacker registers a harmful library using that exact fabricated name



A developer integrates the fake library into their codebase and deploys it

AI may be deceived by fake sites via the Web search tool



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GitHub Copilot Hijacked (CVE-2025-53773, CVSS 7.8):

- Hidden prompt injection tricked the AI into editing local VS Code settings.
- Secretly enabled the `autoApprove` execution flag.
- AI gained silent permission to run any command on the host machine without human consent.

Extending the focus to AI



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- AI agents often come with built-in sandboxing, but not as strong as VM
 - Some AI agents use `sandbox-exec` (similar to Landlock) on macOS, but it has been deprecated since circa 2016
 - Apple recommends using [App Sandbox](#), but not a direct replacement
- Lima can be used as a universal sandbox for any AI agent

AI Inside vs. Outside Lima

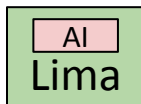


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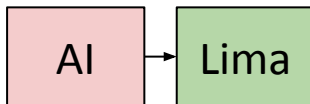
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- **AI inside Lima**

- Just run Codex, Copilot, Claude, Gemini, OpenCode, etc. inside Lima
- LLM inference can be done inside Lima, using GPU acceleration



- **AI outside Lima**

- Lima's MCP server can be connected from AI agents running on the host
- VScode + Remote SSH + Copilot works well too with Lima

Example of AI inside Lima



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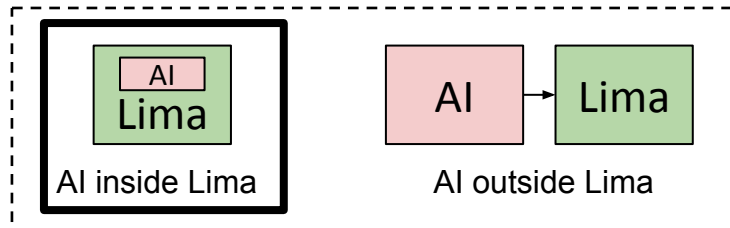
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- Examples in <https://lima-vm.io/docs/examples/ai/>

```
$ limactl start --mount-only .:w  
$ lima claude
```

Only mount the working directory
in read-write mode



GPU acceleration (krunkit)



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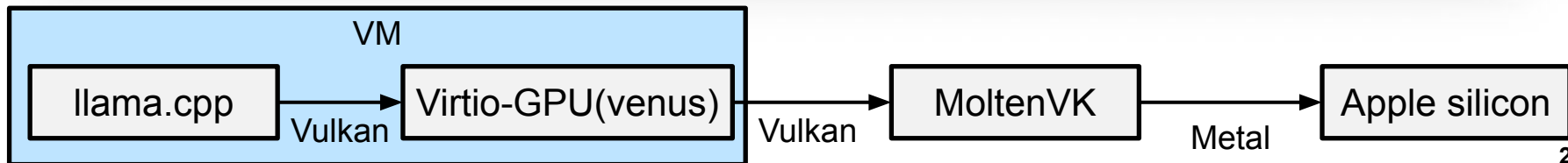


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lima

```
$ lima --version
limactl version 2.0.0
$ lima llama-cli --version
ggml_vulkan: Found 1 Vulkan devices:
ggml_vulkan: 0 = Virtio-GPU Venus (Apple M4 Max) (venus) | uma: 1 | fp16: 1 | bf
16: 0 | warp size: 32 | shared memory: 32768 | int dot: 0 | matrix cores: none
version: 6962 (230d1169e)
built with cc (GCC) 15.2.1 20251022 (Red Hat 15.2.1-3) for aarch64-redhat-linux
$
```



Example of AI outside Lima (MCP Tools)



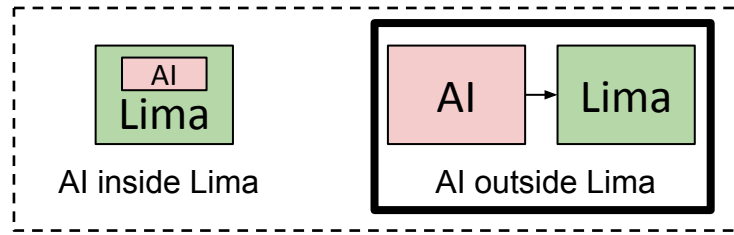
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- Lima exposes several MCP tools for agents running outside VM
 - `list_directory`, `read_file`, `write_file`
 - `run_shell_command`
 - ...
- Similar to Gemini CLI's built-in tools, but strongly sandboxed using VM



- Sync mode
 - Unlike mounts, synced dirs are written back only after user confirmation
 - Prevents AI from “Sorry I removed everything including `.git` dir 🤖”

```
$ limactl start --mount-none
$ lima --sync . claude "Implement something"
[...]
⚠ Accept the changes? (added: 2, deleted: 1, modified: 3, ...)
→ Yes
  No
  View the changed contents
```

v2.1 (March 2026)

- macOS guests

```
$ limctl start template:macos
```

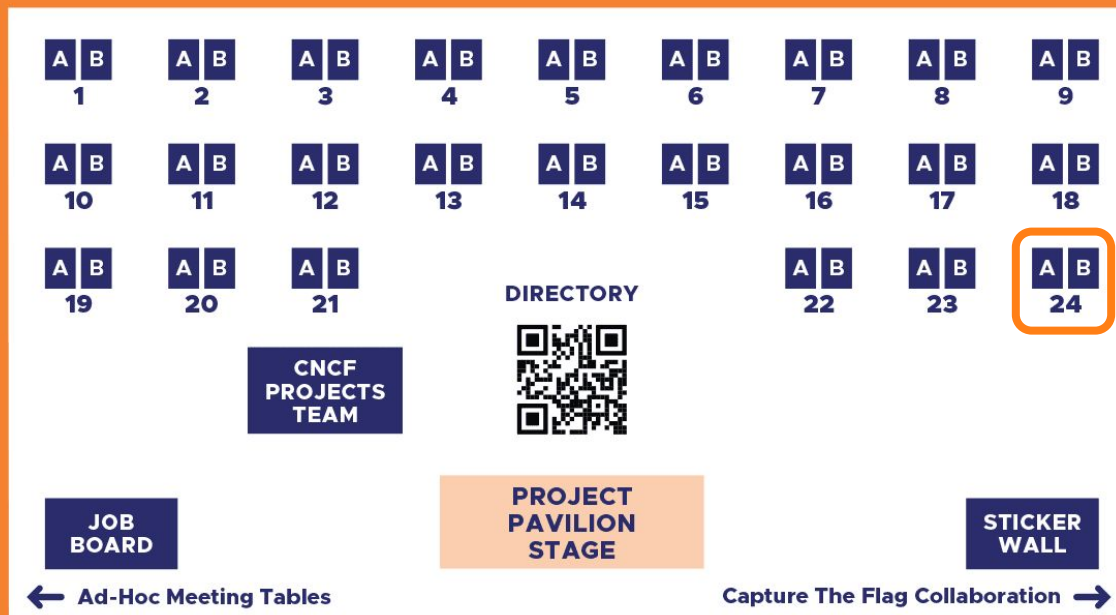
- FreeBSD guests

```
$ limctl start template:freebsd
```

Future ideas

- More VM drivers
 - e.g., for managing IaaS instances
- Windows guests
- Menu-based text user interface
- UX improvement for composing multiple VMs (“Lima Compose”)
- Support for AI workloads on Linux and Windows host

Meet us at the Pavilion today!



14:00 - 17:00

Join our community!



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Web site: <https://lima-vm.io/>

GitHub: <https://github.com/lima-vm/lima>

Slack: <https://slack.cncf.io/> (Channel: #lima)

X (Twitter): [@TheLimaProject](https://twitter.com/TheLimaProject)

Mastodon: [@TheLimaProject@mastodon.social](https://mastodon.social/@TheLimaProject)

Monthly meetings:

<https://zoom-lfx.platform.linuxfoundation.org/meetings/lima>

